

Project Description: Understanding Customer Segment Response to Price Changes for a Product

This project was undertaken with the primary **objective** of understanding how different customer segments respond to price changes for a specific product. This involves analyzing how conversion rates and potentially other behaviors vary across segments (Loyal, New, Returning) when exposed to different price points (High, Low, Medium). The ultimate goal is to inform targeted pricing and marketing strategies.

The analysis began with the fundamental step of **data loading**, which brought in the necessary customer and transaction data for the study.

Following data ingestion, an **exploratory data analysis** provided initial insights into customer behavior:

- **Segment-wise Conversion Summary:** This revealed distinct conversion patterns across different customer segments. Loyal customers exhibited the highest mean conversion rate at 20.02%, followed by Returning customers at 11.53%, and New customers with the lowest at 7.77%. The median conversion rate for all segments was 0.0, which is expected for conversion data where most individual users do not convert (0) and some do (1), leading to many zeros. The user counts for Loyal, New, and Returning segments were 999, 2033, and 1968 respectively.
- **Price Group-wise Conversion Summary:** Analyzing conversion by price group also showed clear differences. The 'Low' price group had the highest mean conversion rate at 16.49%, followed by 'Medium' at 11.16%, and 'High' at 7.34%. Again, median conversion rates were 0.0 across all price groups. The user counts for High, Low, and Medium price groups were 1417, 1522, and 2061 respectively.
- **Visualizations:** To enhance understanding, a **Segment × Price Group Conversion Heatmap** was generated, likely illustrating the interaction between segment and price on conversion. A **Barplot of Conversion Rate per Segment** visually confirmed the segment-wise differences, while a **Boxplot of Purchase Amount by Price Group (Converted Users Only)** provided insights into the transaction value based on price for those who converted.

To statistically confirm the observed relationships, **Independence Tests using Chi-Square** were performed:

- **Segment vs. Conversion:** The Chi-Square test for Segment vs. Converted yielded a Chi2 Statistic of 97.3552 with an associated p-value of 0.0000 (and 2 Degrees of Freedom). The expected frequencies were also provided, showing no immediate concerns for very low expected counts. Given the extremely low p-value, it was concluded that there is a **significant relationship between customer segment and conversion**, meaning conversion rates are not independent of the segment.

- **Price Group vs. Conversion:** Similarly, the Chi-Square test for Price Group vs. Converted resulted in a Chi2 Statistic of 60.4848 with a p-value of 0.0000 (and 2 Degrees of Freedom). The expected frequencies were also presented. This also led to the conclusion that there is a **significant relationship between price group and conversion**, indicating that conversion rates are dependent on the price group.

Further statistical scrutiny involved **Normality and Homogeneity Tests** on the conversion data within price groups, specifically examining the converted variable which is binary (0 or 1).

- **Shapiro-Wilk Test for Normality:** The Shapiro-Wilk test was applied to 'High', 'Low', and 'Medium' price groups. For all three, the W-statistic was low (0.2850, 0.4471, 0.3640 respectively), and the p-values were 0.0000. This strongly indicated that the data in all groups is **not normally distributed**, which is expected for binary (converted/not converted) data. This finding is crucial as it suggests that parametric tests relying on normality assumptions might not be appropriate for analyzing the conversion *variable* itself.
- **Homogeneity of Variances (Levene's Test):** Levene's Test was conducted to assess if the variances of the conversion data were equal across price groups. The test yielded a W-statistic of 30.5943 and a p-value of 0.0000. This result indicated that the **variances differ significantly** across the groups. Both the non-normality and heterogeneity of variances strongly pointed towards the need for non-parametric tests for comparing groups.

Consequently, **Kruskal-Wallis Tests (Non-Parametric)** were performed for both price groups and customer segments to compare their respective distributions (likely of conversion rates or a related continuous metric if converted was transformed, though typically used for non-normal continuous data comparisons).

- **Price Group:** The Kruskal-Wallis Test for price group comparisons yielded an H-statistic of 40.1594 with a p-value of 0.0000. This result confirmed a **significant difference (non-parametric)** in the distributions across price groups.
- **Segment:** The Kruskal-Wallis Test for user segments resulted in an H-statistic of 97.6203 with a p-value of 0.0000. This also confirmed a **significant difference (non-parametric)** in the distributions across user segments.

To understand the practical significance of these statistically significant differences, **Effect Size** was calculated using Epsilon Squared (ϵ^2) for the Kruskal-Wallis tests:

- **Price Group Effect Size:** Epsilon Squared was 0.0076. This was interpreted as a **negligible effect**, indicating that while statistical differences exist, price changes within the tested ranges do not meaningfully influence the observed behavior (e.g., conversion rate or related distribution).
- **Segment Effect Size:** Epsilon Squared was 0.0191. This was interpreted as a **small effect**, suggesting that while the difference in behavior across segments is present and statistically significant, its practical magnitude is modest.

Finally, **Confidence Intervals** were calculated for the conversion rates within each price group and segment:

- **Confidence Interval - Price Group:**
 - High: 7.34% (CI: 5.98% - 8.70%)
 - Low: 16.49% (CI: 14.63% - 18.36%)
 - Medium: 11.16% (CI: 9.80% - 12.52%)
 - **Confidence Interval - Segment:**
 - Loyal: 20.02% (CI: 17.54% - 22.50%)
 - New: 7.77% (CI: 6.61% - 8.94%)
 - Returning: 11.53% (CI: 10.12% - 12.95%)
- These confidence intervals provide a range for the true conversion rates within each category, reinforcing the observed differences. A **Sensitivity visualization** was also included, likely demonstrating how results might change under different parameters or assumptions.
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Summary and Business Decisions

The comprehensive analysis provides a nuanced understanding of customer behavior in response to pricing, differentiating between statistical significance and practical effect.

Key Summary Findings:

1. **Price Impact on Conversion:** Lower prices consistently increase the likelihood of users converting, regardless of their customer segment.
2. **Loyal User Price Sensitivity:** Loyal users exhibit the highest overall conversion rates but also show the steepest decline in conversion when moving from low to high price points. This indicates they are highly price-sensitive despite their overall higher propensity to convert.
3. **Returning User Price Sensitivity:** Returning users display moderate price sensitivity, suggesting they are influenced by price but perhaps not as strongly as loyal customers.
4. **New User Price Sensitivity:** New users appear to be the least sensitive to price. Their generally low conversion rate might be driven more by factors like a lack of trust or familiarity with the product/brand rather than the direct cost.
5. **Price Group Effect Size:** While statistically significant differences were found between price groups (especially noticeable with large sample sizes), the **practical impact (effect size) was negligible**. This implies that simply tweaking prices within the tested ranges may not meaningfully influence overall customer behavior in a way that generates substantial business impact. The implication is to consider investing resources elsewhere.
6. **Segment Effect Size:** The differences in behavior across segments were statistically significant and exhibited a **small but not trivial practical effect size**. This indicates that segment-specific behaviors exist and are worth exploring for targeted strategies.

Actionable Business Decisions:

Based on these findings, the following business decisions are recommended:

1. **Don't Rely on Price Changes Alone:** Given the negligible overall effect size of price on conversion, solely relying on price changes as a primary lever for influencing customer behavior may not be the most effective strategy. Resources might be better allocated elsewhere.
2. **Target Users by Segment:** The observed differences in segment behavior, even if small in overall effect size, are meaningful enough to warrant segment-specific marketing and personalized user experiences.
3. **Careful Pricing for Loyal Users:** Loyal users convert well but are highly price-sensitive. Therefore, pricing strategies for this segment should be carefully managed, potentially leveraging loyalty discounts, exclusive offers, or value-added perks rather than aggressive general price cuts.
4. **Moderate Offers for Returning Users:** Returning users show moderate price sensitivity. Testing moderate offers, such as free shipping or small bundled discounts, could be effective for this segment.
5. **Focus on Trust-Building for New Users:** New users are less price-sensitive but have lower conversion rates, likely due to a lack of trust. Marketing efforts for this segment should focus on building trust through transparent reviews, strong guarantees, excellent customer service, and clear product value propositions, rather than primarily on price.
6. **Invest in Personalization and User Experience:** Overall, the insights suggest that investing in personalization strategies and enhancing the overall user experience will likely yield a higher return on investment than continuous price tweaking across the board.